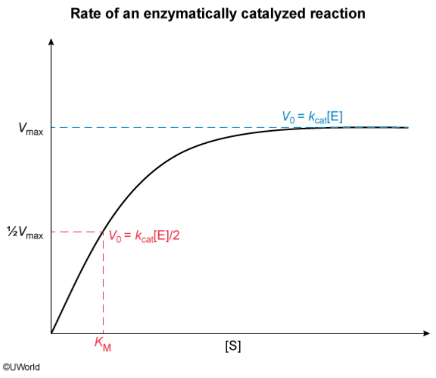
A kinetics experiment reveals that the enzyme fumarase catalyzes the dehydration of malate with a k_{cat} of 900 s^{-1}. If the substrate concentration is equal to the K_M of the reaction, what is the reaction velocity when the enzyme concentration is 200 nM?

- A. 3.6×10^2 μM/s [12%]
- B. 9.0×10^1 μM/s [36%]
- C. 4.5×10^{-3} μM/s [30%]
- D. 2.2×10^{-4} μM/s [20%]

Correct
35% answered correctly

Explanation:



The **rate V_0 of an enzymatically catalyzed reaction** is determined by the enzyme concentration [E], the substrate concentration [S], the Michaelis constant K_M, and the turnover number (reactions per second per enzyme, known as k_{cat}). This relationship is given in the **Michaelis-Menten equation**:

$$V_0 = \frac{k_{cat}[E][S]}{K_M + [S]}$$

The **product of k_{cat} and [E]** equals the **maximum velocity V_{max}** of the reaction, which simplifies the equation to:

$$V_0 = \frac{V_{max}[S]}{K_M + [S]}$$

The question states that the concentration of malate [S] is equal to K_M. Therefore, K_M can be substituted into the equation to yield

$$V_0 = \frac{V_{max} K_M}{K_M + K_M} =$$

Finally, the K_M terms in the numerator and the denominator cancel each other to yield

$$V_0 = \frac{V_{max}}{2}$$

Therefore, when the **substrate concentration equals the K_M** of the catalyzed reaction, the reaction will proceed at half the maximum possible velocity, or **½V_{max}**. The question says the concentration of fumarase [E] is 200 nM, and the k_{cat} is 900 s^{-1}. Multiplying these terms yields a V_{max} of 180,000 nM/s, which is equivalent to 180 μM/s (1,000 nM = 1 μM). Per the derived equation, dividing V_{max} by 2 yields a reaction rate V_0 of 90 μM/s, which is equal to **9.0×10^1 μM/s**.

(Choice A) 3.6×10^2 μM/s is double the maximum velocity V_{max} of the reaction at the given enzyme concentration. The question states that the concentration of malate is equal to the K_M of fumarase, so the reaction will proceed at ½V_{max}.

(Choice C) 4.5×10^{-3} μM/s results from incorrectly calculating V_{max} by *dividing* k_{cat} by the fumarase concentration [E] instead of multiplying.

(Choice D) 2.2×10^{-4} μM/s results from incorrectly calculating V_{max} by *dividing* the fumarase concentration [E] by k_{cat} instead of multiplying.

Educational objective:
The rate of an enzymatic reaction at any given substrate concentration is described by the Michaelis-Menten equation. The maximum velocity V_{max} equals the product of k_{cat} and enzyme concentration [E]. When the substrate concentration equals the Michaelis constant K_M, the reaction proceeds at $\frac{1}{2}V_{max}$.

Time spent: 0 sec	**QID**:402030
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